

C PARTICULAR CASE: ISOTROPIC COVARIANCE MATRIX

Here, we consider a simple covariance matrix of the form $\mathbf{C} = \sigma^2 \mathbf{I}_p$ for some $\sigma > 0$. So

$$\delta_S = \alpha \sigma^2 \delta \quad (21)$$

C.1 RESOLVENT IDENTITIES IN THE CASE OF $\mathbf{C} = \sigma^2 \mathbf{I}_p$

We have that by lemma A.4:

$$\begin{aligned} \bar{\mathbf{Q}} &= \left(\frac{\pi}{1+\delta} (\boldsymbol{\mu} \boldsymbol{\mu}^\top + \mathbf{I}_p) + \frac{\alpha(1-\pi)}{1+\alpha\sigma^2\delta} (\boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \sigma^2 \mathbf{I}_p) + \gamma \mathbf{I}_p \right)^{-1} \\ &= \left(\frac{\pi}{1+\delta} \boldsymbol{\mu} \boldsymbol{\mu}^\top + \frac{\alpha(1-\pi)}{1+\alpha\sigma^2\delta} \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \theta \mathbf{I}_p \right)^{-1}, \end{aligned}$$

where:

$$\theta = \gamma + \frac{\pi}{1+\delta} + \frac{\alpha\sigma^2(1-\pi)}{1+\alpha\sigma^2\delta} \quad (22)$$

Define by:

$$\mathbf{R}_1 = \left(\frac{\alpha(1-\pi)}{1+\alpha\sigma^2\delta} \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \theta \mathbf{I}_p \right)^{-1}, \quad \mathbf{R}_2 = \left(\frac{\pi}{1+\delta} \boldsymbol{\mu} \boldsymbol{\mu}^\top + \theta \mathbf{I}_p \right)^{-1}, \quad (23)$$

such that:

$$\bar{\mathbf{Q}} = \left(\frac{\pi}{1+\delta} \boldsymbol{\mu} \boldsymbol{\mu}^\top + \mathbf{R}_1^{-1} \right)^{-1} = \left(\frac{\alpha(1-\pi)}{1+\alpha\sigma^2\delta} \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \mathbf{R}_2^{-1} \right)^{-1}$$

Thus, using lemma A.3:

$$\bar{\mathbf{Q}} \boldsymbol{\mu} = \frac{\mathbf{R}_1 \boldsymbol{\mu}}{1 + \frac{\pi}{1+\delta} \boldsymbol{\mu}^\top \mathbf{R}_1 \boldsymbol{\mu}}, \quad \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta = \frac{\mathbf{R}_2 \boldsymbol{\mu}_\beta}{1 + \frac{\alpha(1-\pi)}{1+\alpha\sigma^2\delta} \boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta} \quad (24)$$

And:

$$\bar{\mathbf{Q}} = \mathbf{R}_1 - \frac{\pi \mathbf{R}_1 \boldsymbol{\mu} \boldsymbol{\mu}^\top \mathbf{R}_1}{1 + \delta + \pi \boldsymbol{\mu}^\top \mathbf{R}_1 \boldsymbol{\mu}} = \mathbf{R}_2 - \frac{\alpha(1-\pi) \mathbf{R}_2 \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top \mathbf{R}_2}{1 + \alpha\sigma^2\delta + \alpha(1-\pi) \boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta} \quad (25)$$

Lemma C.1 (Delta). *The parameter δ as defined in equation 9, is given by the following identity:*

$$\delta = \frac{\eta}{\theta} = \frac{\eta}{\gamma + \frac{\pi}{1+\delta} + \frac{\alpha\sigma^2(1-\pi)}{1+\alpha\sigma^2\delta}}$$

Which gives us a third order equation:

$$\alpha\sigma^2\gamma\delta^3 + (\gamma + \alpha\sigma^2(1 + \gamma - \eta))\delta^2 + (\gamma + \pi - \eta + \alpha\sigma^2(1 - \pi - \eta))\delta - \eta = 0$$

Lemma C.2 (Resolvent identities). *Using the first identity in Sherman-Morrisson's lemma A.3, we have that the expressions of $\mathbf{R}_1$ and $\mathbf{R}_2$ are given by:*

$$\mathbf{R}_1 = \frac{1}{\theta} \mathbf{I}_p - \frac{\alpha(1-\pi) \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top}{\theta^2(1 + \alpha\sigma^2\delta) + \theta\alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2}, \quad \mathbf{R}_2 = \frac{1}{\theta} \mathbf{I}_p - \frac{\pi \boldsymbol{\mu} \boldsymbol{\mu}^\top}{\theta^2(1 + \delta) + \theta\pi\|\boldsymbol{\mu}\|^2}$$

And we also have the following identities:

$$\mathbf{R}_1 \boldsymbol{\mu}_\beta = \frac{\boldsymbol{\mu}_\beta}{\theta + \frac{\alpha(1-\pi)}{1+\delta_S} \|\boldsymbol{\mu}_\beta\|^2}, \quad \mathbf{R}_2 \boldsymbol{\mu} = \frac{\boldsymbol{\mu}}{\theta + \frac{\pi}{1+\delta} \|\boldsymbol{\mu}\|^2}$$

$$\boldsymbol{\mu}^\top \mathbf{R}_1 \boldsymbol{\mu} = \frac{\|\boldsymbol{\mu}\|^2}{\theta} \left(1 - \frac{\alpha(1-\pi)\beta^2 \|\boldsymbol{\mu}\|^2}{\theta(1 + \delta_S) + \alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2} \right) = \frac{\|\boldsymbol{\mu}\|^2}{\theta} \frac{\theta(1 + \delta_S) + \alpha(1-\pi)(1 - \beta^2) \|\boldsymbol{\mu}^\perp\|^2}{\theta(1 + \delta_S) + \alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2}$$

$$\boldsymbol{\mu}^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta = \frac{\beta(1 + \delta) \|\boldsymbol{\mu}\|^2}{\theta(1 + \delta) + \pi \|\boldsymbol{\mu}\|^2}$$

$$\begin{aligned}
\boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta &= \frac{1}{\theta} \left(\|\boldsymbol{\mu}_\beta\|^2 - \frac{\pi \beta^2 \|\boldsymbol{\mu}\|^4}{\theta(1+\delta) + \pi \|\boldsymbol{\mu}\|^2} \right) \\
\boldsymbol{\mu}^\top \mathbf{R}_1^2 \boldsymbol{\mu} &= \frac{\|\boldsymbol{\mu}\|^2}{\theta^2} \left(1 - \frac{2\alpha(1-\pi)\beta^2 \|\boldsymbol{\mu}\|^2}{\theta(1+\delta_S) + \alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2} + \frac{\alpha^2(1-\pi)^2 \beta^2 \|\boldsymbol{\mu}\|^2 \|\boldsymbol{\mu}_\beta\|^2}{(\theta(1+\delta_S) + \alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2)^2} \right) \\
&= \frac{\|\boldsymbol{\mu}\|^2}{\theta^2} + \frac{\alpha(1-\pi)\beta^2 \|\boldsymbol{\mu}\|^4}{\theta^2(\theta(1+\delta_S) + \alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2)} \left(\frac{\alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2}{\theta(1+\delta_S) + \alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2} - 2 \right) \\
\boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \mathbf{R}_1 \boldsymbol{\mu} &= \frac{\beta \|\boldsymbol{\mu}\|^2}{\theta^2} \left(1 - \frac{\alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2}{\theta(1+\delta_S) + \alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2} - \frac{\pi \|\boldsymbol{\mu}\|^2}{\theta(1+\delta) + \pi \|\boldsymbol{\mu}\|^2} \right. \\
&\quad \left. + \frac{\alpha\pi(1-\pi)\beta^2 \|\boldsymbol{\mu}\|^4}{(\theta(1+\delta) + \pi \|\boldsymbol{\mu}\|^2)(\theta(1+\delta_S) + \alpha(1-\pi)\|\boldsymbol{\mu}_\beta\|^2)} \right) \\
\boldsymbol{\mu}_\beta^\top \mathbf{R}_2^2 \boldsymbol{\mu}_\beta &= \frac{\|\boldsymbol{\mu}_\beta\|^2}{\theta^2} + \frac{\pi \beta^2 \|\boldsymbol{\mu}\|^4}{\theta^2(\theta(1+\delta) + \pi \|\boldsymbol{\mu}\|^2)} \left(\frac{\pi \|\boldsymbol{\mu}\|^2}{\theta(1+\delta) + \pi \|\boldsymbol{\mu}\|^2} - 2 \right) \\
&= \frac{\|\boldsymbol{\mu}_\beta\|^2}{\theta^2} - \frac{\pi \beta^2 \|\boldsymbol{\mu}\|^4 (\pi \|\boldsymbol{\mu}\|^2 + 2\theta(1+\delta))}{\theta^2(\theta(1+\delta) + \pi \|\boldsymbol{\mu}\|^2)^2}
\end{aligned}$$

Lemma C.3 (Trace identities). *Let $i \in \{1, \dots, n\}$, and $j \in \{1, \dots, m\}$, such that: $\Sigma = \mathbb{E}[\mathbf{x}_i \mathbf{x}_i^\top] = \boldsymbol{\mu} \boldsymbol{\mu}^\top + \mathbf{I}_p$ and $\Sigma_\beta = \mathbb{E}[\tilde{\mathbf{x}}_j \tilde{\mathbf{x}}_j^\top] = \boldsymbol{\mu}_\beta \boldsymbol{\mu}_\beta^\top + \sigma^2 \mathbf{I}_p$.*

We can prove that:

$$\frac{1}{N} \text{Tr}((\Sigma \bar{\mathbf{Q}})^2) = \frac{\eta}{\theta^2}, \quad \frac{1}{N} \text{Tr}((\Sigma_\beta \bar{\mathbf{Q}})^2) = \frac{\eta \sigma^4}{\theta^2}, \quad \frac{1}{N} \text{Tr}(\Sigma_\beta \bar{\mathbf{Q}} \Sigma \bar{\mathbf{Q}}) = \frac{\eta \sigma^2}{\theta^2}$$

The performance of $\mathbf{w}_q$ in (6) is fully determined by the first two order moments: $\mathbb{E}[\mathbf{w}_q^\top \mathbf{x}]$ and $\mathbb{E}[(\mathbf{w}_q^\top \mathbf{x})^2]$.

C.2 TEST EXPECTATION

We have that using the calculus in the past section:

$$\mathbb{E}[\mathbf{w}_q^\top \mathbf{x}] = (-1)^a \left(\frac{\pi}{1+\delta} \boldsymbol{\mu}^\top + \frac{\lambda(1-\pi)}{1+\alpha\sigma^2\delta} \boldsymbol{\mu}_\beta^\top \right) \bar{\mathbf{Q}} \boldsymbol{\mu} \quad (26)$$

And finally we use lemma C.2 and the following identities to obtain the result:

$$\boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} = \frac{\boldsymbol{\mu}^\top \mathbf{R}_1 \boldsymbol{\mu}}{1 + \frac{\pi}{1+\delta} \boldsymbol{\mu}^\top \mathbf{R}_1 \boldsymbol{\mu}}, \quad \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu} = \frac{\boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}}{1 + \frac{\alpha(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta}$$

C.3 TEST VARIANCE

To determine the variance of $\mathbf{w}_q^\top \mathbf{x}$, it only remains to compute its second order. We have that:

$$\begin{aligned}
\mathbb{E}[(\mathbf{w}_q^\top \mathbf{x})^2] &= \frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} + \sum_{j=1}^m q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x} \right)^2 \right] \\
&= \frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \right)^2 \right] + \frac{1}{N^2} \mathbb{E} \left[\left(\sum_{j=1}^m q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x} \right)^2 \right] + \frac{2}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \right) \left(\sum_{j=1}^m q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x} \right) \right]
\end{aligned}$$

And using the same computations in the past section, we get:

First sum: The first sum gives us:

$$\frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^n y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} \right)^2 \right] = \frac{\pi^2}{(1+\delta)^2} \left(\boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} - \frac{2 \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])}{N(1+\delta)} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \right) + \frac{\pi}{N(1+\delta)^2} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])$$

Second sum: the second sum gives us:

$$\begin{aligned} & \frac{1}{N^2} \mathbb{E} \left[\left(\sum_{i=1}^m q_i \tilde{y}_i \tilde{\mathbf{x}}_i^\top \mathbf{Q} \mathbf{x} \right)^2 \right] \\ &= \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \left(\boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta - \frac{2}{N(1+\delta_S)} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) + \frac{\alpha(1-\pi)}{N(1+\delta_S)^2} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) \end{aligned}$$

Third sum: The third sum is given by:

$$\begin{aligned} & \frac{2}{N^2} \sum_{i=1}^n \sum_{j=1}^m \mathbb{E}[y_i \mathbf{x}_i^\top \mathbf{Q} \mathbf{x} q_j \tilde{y}_j \tilde{\mathbf{x}}_j^\top \mathbf{Q} \mathbf{x}] \\ &= \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)} \left(\boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta - \frac{1}{N} \text{Tr} \left(\left(\frac{\Sigma}{1+\delta} + \frac{\Sigma_\beta}{1+\delta_S} \right) \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \right) \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \end{aligned}$$

Grouping all the sums: Denote by : $T = \frac{1}{N} \text{Tr}(\Sigma \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}])$, then: $\frac{1}{N} \text{Tr}(\Sigma_\beta \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}]) = \sigma^2 T$. Now let us group the terms in T in the three sums, and those that do not depend on T . We get that:

$$\begin{aligned} \mathbb{E}[(\mathbf{w}^\top \mathbf{x})^2] &= \frac{\pi^2}{(1+\delta)^2} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} + \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta + \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta \\ &+ T \left(\frac{\pi}{(1+\delta)^2} - \frac{2\pi^2}{(1+\delta)^3} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} - \frac{2\lambda\pi(1-\pi)}{(1+\delta)^2(1+\delta_S)} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \\ &+ \sigma^2 T \left(\frac{\alpha(1-\pi)}{(1+\delta_S)^2} - \frac{2\lambda^2(1-\pi)^2}{(1+\delta_S)^3} \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta - \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)^2} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \\ &= \frac{\pi^2}{(1+\delta)^2} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} + \frac{\lambda^2(1-\pi)^2}{(1+\delta_S)^2} \boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta + \frac{2\lambda\pi(1-\pi)}{(1+\delta)(1+\delta_S)} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta \\ &+ \frac{\pi T}{(1+\delta)^2} \left(1 - \frac{2\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} - \frac{2\lambda(1-\pi)}{1+\delta_S} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \\ &+ \frac{(1-\pi)\sigma^2 T}{(1+\delta_S)^2} \left(\alpha - \frac{2\lambda^2(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta - \frac{2\lambda\pi}{1+\delta} \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \right) \end{aligned}$$

And we can compute this since we have that:

$$\begin{aligned} \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu} &= \frac{1}{h} ((1-b_2) ((\boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu})^2 + \boldsymbol{\mu}^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu}) + b_1 ((\boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta)^2 + \sigma^2 \boldsymbol{\mu}^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu})), \\ \boldsymbol{\mu}_\beta^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta &= \frac{1}{h} ((1-b_2) [(\boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu})^2 + \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu}_\beta] + b_1 [(\boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta)^2 + \sigma^2 \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu}_\beta]) \\ \boldsymbol{\mu}^\top \mathbb{E}[\mathbf{Q} \Sigma \mathbf{Q}] \boldsymbol{\mu}_\beta &= \frac{1}{h} ((1-b_2) [\boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu} \cdot \boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta + \boldsymbol{\mu}^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu}_\beta] + b_1 [\boldsymbol{\mu}^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta \cdot \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta + \sigma^2 \boldsymbol{\mu}^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu}_\beta]) \\ \boldsymbol{\mu}_\beta \bar{\mathbf{Q}} \boldsymbol{\mu}_\beta &= \frac{\boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta}{1 + \frac{\alpha(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta}, \quad \boldsymbol{\mu}^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu} = \frac{\boldsymbol{\mu}^\top \mathbf{R}_1^2 \boldsymbol{\mu}}{\left(1 + \frac{\pi}{1+\delta} \boldsymbol{\mu}^\top \mathbf{R}_1 \boldsymbol{\mu}\right)^2}, \quad \boldsymbol{\mu}_\beta^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu}_\beta = \frac{\boldsymbol{\mu}_\beta^\top \mathbf{R}_2^2 \boldsymbol{\mu}_\beta}{\left(1 + \frac{\alpha(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta\right)^2}, \\ \boldsymbol{\mu}^\top \bar{\mathbf{Q}}^2 \boldsymbol{\mu}_\beta &= \frac{\boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \mathbf{R}_1 \boldsymbol{\mu}}{\left(1 + \frac{\pi}{1+\delta} \boldsymbol{\mu}^\top \mathbf{R}_1 \boldsymbol{\mu}\right) \left(1 + \frac{\alpha(1-\pi)}{1+\delta_S} \boldsymbol{\mu}_\beta^\top \mathbf{R}_2 \boldsymbol{\mu}_\beta\right)} \end{aligned}$$

Theorem C.4 (Gaussianity of the 6 model for $\mathbf{C} = \sigma^2 \mathbf{I}_p$). *Let $\mathbf{w}_q$ be the Mixed classifier as defined in equation 6 and suppose that Assumption 4.1 holds. The decision function $\mathbf{w}_q^\top \mathbf{x}$, on some test sample $\mathbf{x} \in \mathcal{C}_a$ independent of $\mathbf{X}$, satisfies:*

$$\mathbf{w}_q^\top \mathbf{x} \xrightarrow{\mathcal{D}} \mathcal{N}((-1)^a m_q, \nu_q - m_q^2),$$